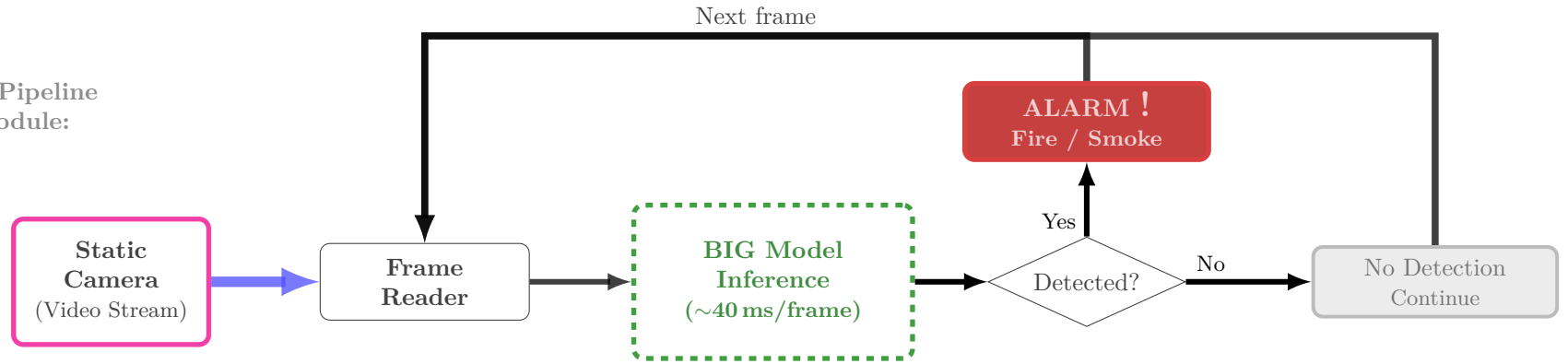
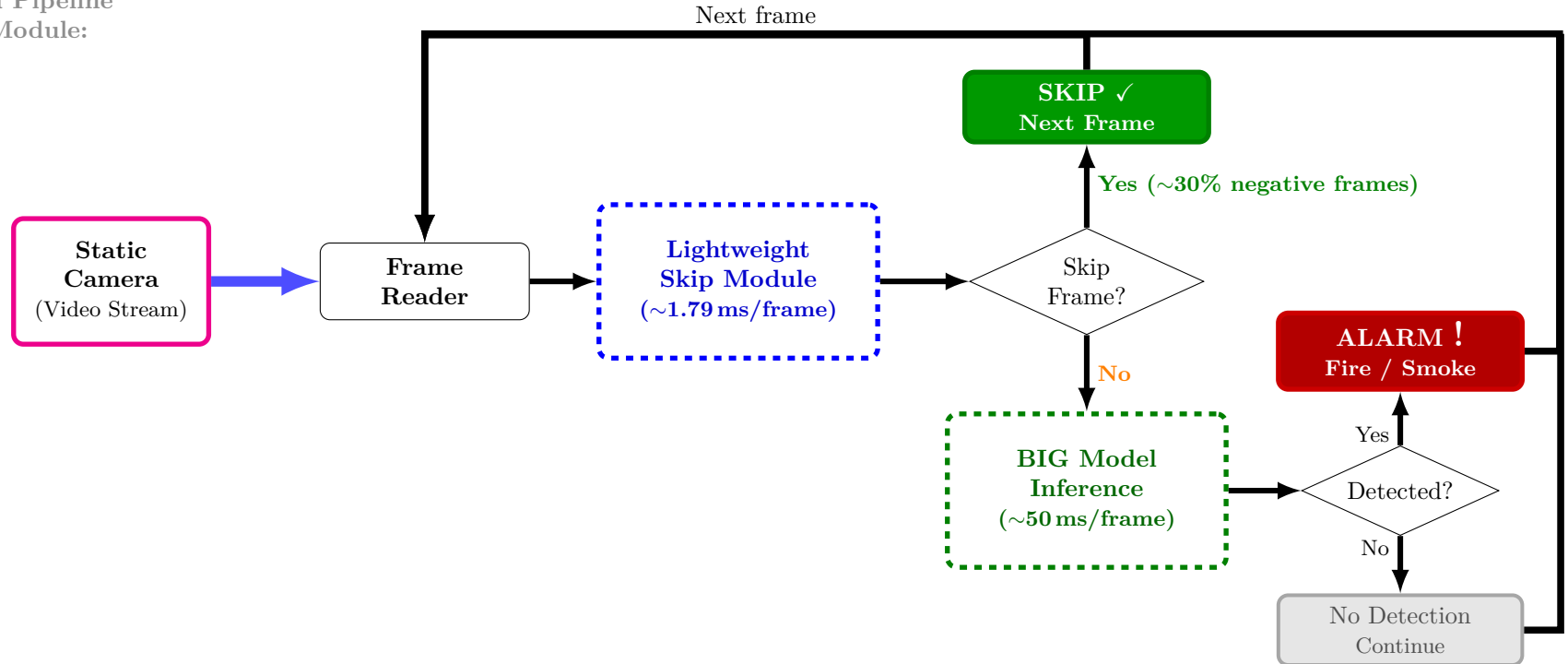


(a) Baseline Pipeline
- No Skip Module:



50 ms/frame — 25 FPS (runs on *every* frame)

(b) Proposed Pipeline
-With Skip Module:



Baseline (no skip): 40 ms/frame (25 FPS) — BIG Model runs on *every* frame

Proposed (with skip): $= (0.27 \times 1.79) + (0.73 \times 41.79) \approx 31.0 \text{ ms/frame } (\approx 32 \text{ FPS})$ **Speedup:** $\approx 1.3\times$ (+29% FPS)

Assumes 90% of all frames are negative, the skip rate of negative frames is 30%.